

IEEE CONFERENCE PAPER PRESENTATION

Behavioral Feature Engineering and Explainable Gradient Boosting for Credit Card Default Prediction

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Summer Internship Project

INTRODUCTION

Understanding Credit Card Default



A Persistent Financial Risk

Credit card default occurs when a cardholder fails to make minimum payments for an extended period. It drives billions in annual losses for issuers and directly shapes interest rates, credit limits, and lending policy.



From Rule-Based Scoring to Machine Learning

Early credit scoring relied on static, rule-based systems like FICO. Modern issuers increasingly use machine learning to capture complex, nonlinear patterns in spending and repayment behavior that fixed rules miss.



Why This Study

This work asks two questions together: can engineered behavioral features improve prediction beyond raw account data, and can the resulting model still be explained clearly enough for regulated lending decisions?

OVERVIEW

What We'll Cover



Motivation & Problem

Why default prediction is hard, and why behavioral signal matters



Dataset & Feature Engineering

30,000 records, 23 original attributes, 10 new behavioral features



Methodology

Six models, five-fold cross-validation, Optuna tuning



Results

Baseline comparison, tuned model performance, robustness checks



Ablation & Significance

Isolating the engineered features with a paired significance test



Explainability & Conclusion

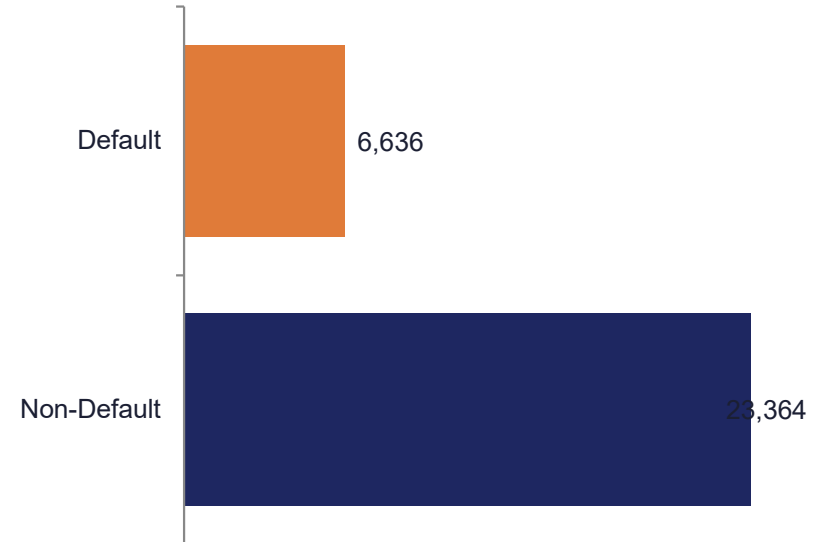
Feature importance, key findings, and future work

MOTIVATION

Why Credit Card Default Prediction Is Hard

- Financial institutions rely on accurate default prediction for lending decisions, portfolio management, and financial stability.
- Traditional scoring (logistic regression) assumes linear relationships and misses evolving repayment behavior.
- Gradient boosting models capture nonlinear patterns well — but most prior work compares algorithms without designing behaviorally meaningful features.

Class Distribution — UCI Credit Card Default Dataset (n = 30,000)



77.9% non-default vs. 22.1% default — moderate class imbalance

BACKGROUND

Related Work



Gradient Boosting Dominance

XGBoost, LightGBM, and CatBoost consistently outperform linear models on structured credit data; CatBoost handles categorical features especially well.



Behavioral Feature Engineering

Repayment behavior observed over time (e.g., mobile-usage or repayment consistency patterns) reveals risk signal invisible in single-period snapshots.



Explainable AI (SHAP)

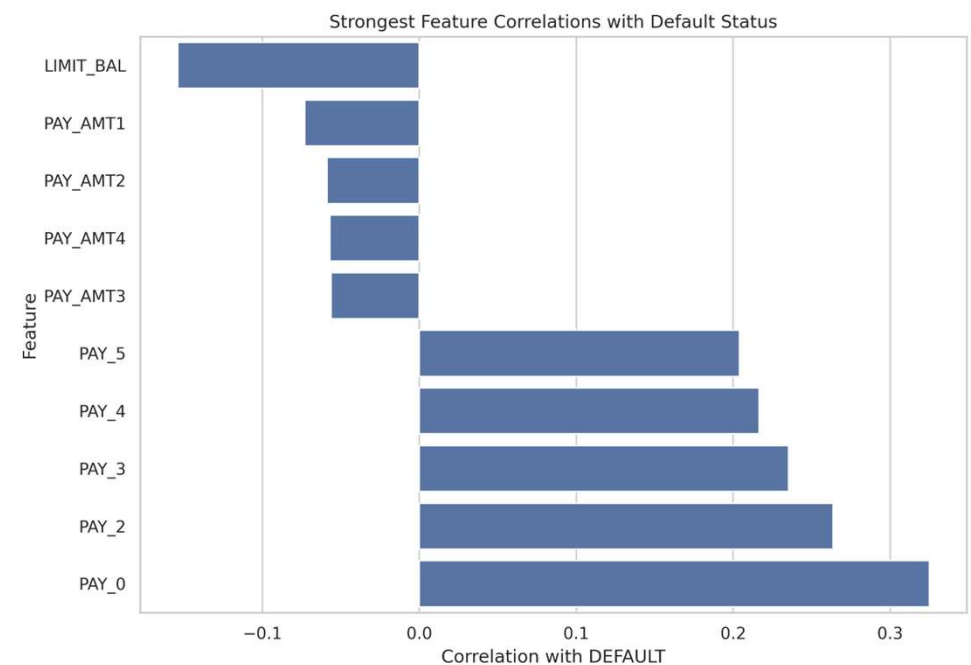
SHAP's TreeExplainer gives exact, efficient Shapley-value computation for tree ensembles — essential for regulated financial deployment.

SECTION 03 • DATASET

Dataset Overview

Statistic	Value
Total records	30,000
Original attributes	23
Non-default cases	23,364 (77.9%)
Default cases	6,636 (22.1%)
Training set (80%, stratified)	24,000
Testing set (20%, stratified)	6,000

Outlier rate: 8–10% of bill/payment values exceed $1.5 \times \text{IQR}$ — retained, since extreme balances are informative of risk, not data errors.



Raw repayment-status columns (PAY_0–PAY_5) are the strongest correlates of default — motivating the engineered features on the next slide.

SECTION 04 • METHODOLOGY

Behavioral Feature Engineering

10 new features appended to the 23 original attributes → 33-feature representation

Repayment Consistency

- `AVG_PAY_STATUS`
- `MAX_PAY_STATUS`
- `DELAYED_MONTHS`
- `SEVERE_DELAY_MONTHS`

Balance & Utilization

- `AVG_BILL_AMT`
- `TOTAL_BILL`
- `CREDIT_UTILIZATION`

Payment Behavior

- `AVG_PAY_AMT`
- `TOTAL_PAYMENT`
- `PAYMENT_RATIO`

Two new counts — `DELAYED_MONTHS` and `SEVERE_DELAY_MONTHS` — capture how often, not just how severely, a customer fell behind.

Project Work

Stage 1 — Core Features (8)

- AVG_PAY_STATUS
- MAX_PAY_STATUS
- AVG_BILL_AMT
- TOTAL_BILL
- CREDIT_UTILIZATION
- AVG_PAY_AMT
- TOTAL_PAYMENT
- PAYMENT_RATIO

Stage 2 — Delay-Frequency Additions (+2)

2 entirely new features were added in the final notebook:

DELAYED_MONTHS — counts how many of the 6 months had *any* payment delay

SEVERE_DELAY_MONTHS — counts how many months had delays of 2+ months (more severe risk signal)

CREDIT_UTILIZATION formula was improved — V1 used a simple division, while the final version handles LIMIT_BAL = 0 by replacing with np.nan to avoid division errors.

PAYMENT_RATIO formula changed — V1 used TOTAL_PAYMENT / (abs(TOTAL_BILL) + 1), while the final version uses AVG_PAY_AMT / AVG_BILL_AMT (a cleaner ratio of averages, also with np.nan and inf handling).

...

	Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
0	Logistic Regression	0.7788	0.0000	0.0000	0.0000	0.6329
1	Decision Tree	0.7312	0.3983	0.4220	0.4098	0.6217
2	Random Forest	0.8118	0.6296	0.3625	0.4601	0.7573
3	XGBoost	0.8198	0.6694	0.3662	0.4735	0.7795
4	LightGBM	0.8188	0.6667	0.3617	0.4690	0.7746
5	CatBoost	0.8168	0.6570	0.3595	0.4647	0.7814

...

	Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
0	CatBoost	0.820874	0.672740	0.371609	0.478640	0.788024
1	Optimized CatBoost	0.818955	0.670922	0.356712	0.465780	0.777994
2	XGBoost	0.818955	0.661727	0.372551	0.476572	0.781094
3	LightGBM	0.818872	0.663655	0.368404	0.473672	0.780570
4	LightGBM + Optuna	0.818288	0.670504	0.351433	0.461158	0.778452
5	Random Forest	0.813866	0.634616	0.374437	0.470921	0.769587
6	Decision Tree	0.812615	0.636476	0.358039	0.457663	0.745808
7	Logistic Regression	0.811697	0.667254	0.297135	0.411132	0.764993

FINAL MODEL COMPARISON TABLE

End-to-End Methodology



All models evaluated with accuracy, precision, recall, F1-score, and ROC-AUC; fold-level standard deviation reported for stability.

Baseline Model Comparison (5-Fold CV)



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FINAL MODEL COMPARISON TABLE

CatBoost — Top Performer

Accuracy	82.09%
Precision	0.6727
Recall	0.3716
F1-Score	0.4786
ROC-AUC	0.7880

Gradient boosting methods (CatBoost, XGBoost, LightGBM) form a clear top cluster. The single Decision Tree trails by a wide margin, as expected from its high variance.

SECTION 06C • ROBUSTNESS

Cross-Validation Robustness Check

Model	Mean ROC-AUC	Std
CatBoost	0.7880	0.0087
XGBoost	0.7811	0.0079
LightGBM	0.7806	0.0096
Random Forest	0.7696	0.0066
Logistic Regression	0.7650	0.0090
Decision Tree	0.7458	0.0103

Low ROC-AUC standard deviation for CatBoost and XGBoost confirms the ranking is stable across folds — not an artifact of one split.

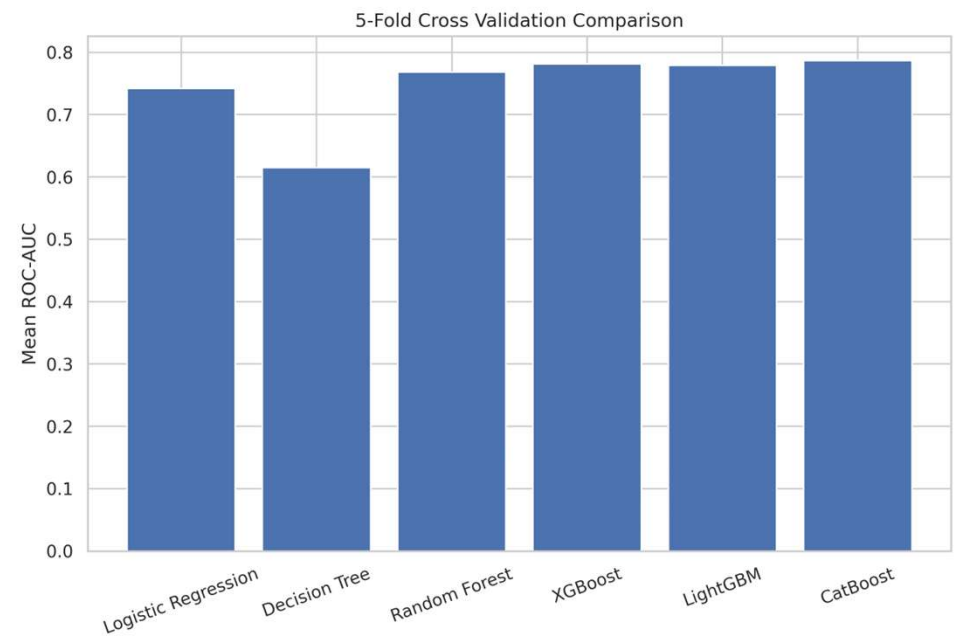


Fig. Cross-validation ROC-AUC distribution across models

SECTION 06B • FINAL MODEL

Final Optuna-Tuned CatBoost Model

Held-out test set (6,000 records) · Decision threshold = 0.50

81.90%

Accuracy

0.6709

Precision

0.3567

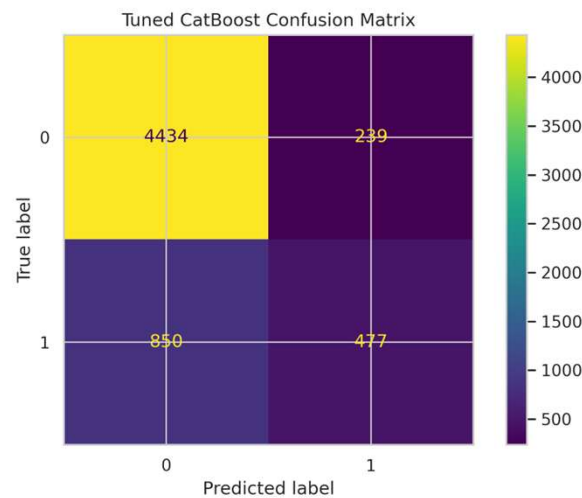
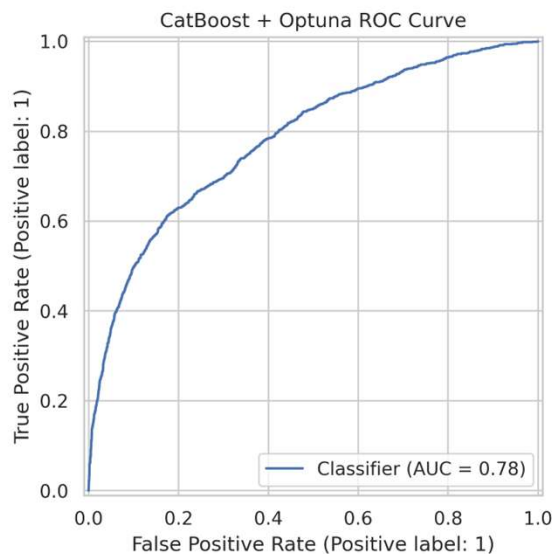
Recall

0.4658

F1-Score

0.7780

ROC-AUC



Reading the Result

- Test ROC-AUC (0.7780) is slightly below the 5-fold CV mean (0.7880) for baseline CatBoost — expected sampling variance between one split and a cross-validated estimate.

Precision/recall reflect the difficulty of the minority default class at a fixed 0.5 threshold.

SECTION 06D • ABLATION STUDY

Isolating the Engineered Features

Feature Set	ROC-AUC	Accuracy	F1
Original (23)	0.7763	0.8188	0.4708
Original + Engineered (33)	0.7780	0.8190	0.4658

Single held-out split

Fold	Original	Engineered	Δ
1	0.7899	0.7922	+0.0022
2	0.7883	0.7913	+0.0030
3	0.7820	0.7817	-0.0003
4	0.7724	0.7749	+0.0026
5	0.7971	0.8000	+0.0029
Mean	0.7859	0.7880	+0.0021

5-fold cross-validation — improves in 4 of 5 folds



Statistical Significance

Wilcoxon signed-rank test

H1: Engineered > Original · one-sided

14.0

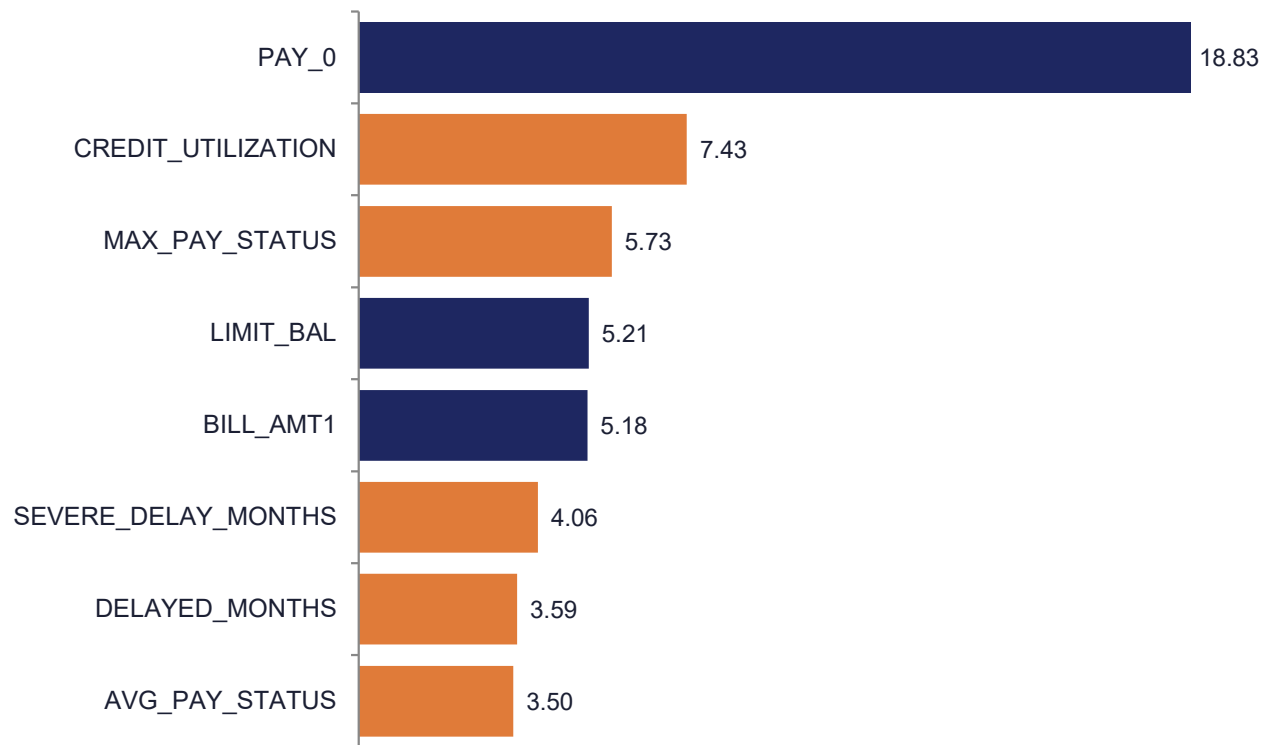
test statistic

p = 0.0625

not significant at $\alpha = 0.05$

Honest takeaway: the improvement is small, directionally consistent, but not statistically decisive at $n = 5$ folds — much of the behavioral signal already exists in the raw PAY_ / BILL_AMT* sequences.*

What Drives the Model's Predictions?



Engineered feature
Original attribute

5 of 8

top predictors are engineered behavioral features

PAY_0 remains the single strongest signal — the most recent repayment status is the most direct indicator of imminent default.

APPLICATIONS

Real-World Applications & Impact



Credit Card Issuers

Banks and card networks screen applications and flag at-risk accounts early, enabling proactive outreach before default occurs.



Regulatory Compliance

SHAP-backed explainability helps models meet fair-lending and audit requirements, where every adverse credit decision must be justifiable.



Fintech & Digital Lenders

Alternative and digital-first lenders apply similar behavioral scoring to underwrite thin-file or newly banked customers.



Portfolio Risk Management

Predicted default probabilities feed into loss forecasting, capital reserve planning, and portfolio-level stress testing.

SUMMARY

Key Findings



CatBoost wins consistently

Best model across every metric except precision — mean CV ROC-AUC 0.7880, confirmed stable across folds.



Engineered features help, modestly

+0.0021 mean CV ROC-AUC (4 of 5 folds); Wilcoxon $p = 0.0625$ — directionally consistent, not yet significant.



Behavioral signal is real, if overlapping

5 of top 8 predictors are engineered — but they partly duplicate signal CatBoost already extracts from raw sequences.



Recall remains the open problem

Recall of 0.36–0.37 at the default 0.5 threshold shows the minority default class is still hard to catch.

LIMITATIONS

Limitations & Threats to Validity

Methodological limitations

- Headline test metrics come from a single 80/20 split — only the ablation study used 5-fold cross-validation
- The Wilcoxon significance test rests on just 5 folds, a small sample for a paired statistical comparison
- A fixed 0.50 decision threshold was used throughout, rather than one tuned to the real cost of a missed default
- The dataset reflects one market and time period (Taiwan, 2005) and may not generalize to other regions or eras

Practical constraints

- Recall of 0.36–0.37 means roughly six in ten true defaulters are still missed at the chosen threshold
- Engineered features partly duplicate signal CatBoost already extracts from raw PAY_* and BILL_AMT* sequences
- SHAP values explain correlations the model learned, not the causal drivers of a customer's default
- No fairness or disparate-impact audit across demographic groups was performed in this study

CONCLUSION

Conclusion & Future Work

What we showed

- A 33-feature behavioral representation (23 original + 10 engineered) for credit default prediction
- CatBoost is the strongest of six models, robust across 5-fold CV (ROC-AUC 0.7880)
- Optuna tuning reaches a held-out test ROC-AUC of 0.7780
- A rigorous ablation with a paired significance test — not just a point estimate

What's next

- Repeated / 10-fold CV for greater statistical power on the ablation result
- Class-imbalance-aware training to improve recall (currently 0.36–0.37)
- Trajectory-based features (fitted slopes) rather than simple aggregates
- Fairness audit across demographic attributes for responsible lending

Thank You

Questions & Discussion

Github Link: <https://github.com/anshulkanswal6-dev/JSS-x-AIHI-Internship-Work/tree/main>